

ENGR 102 Summer Bridge**Day 5 Student Quiz**

Compound Boolean Evidence and Complete Decisions

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as **and**, **or**, and **not**. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Compare two Boolean groupings.**5 points**

```
trained = False
temperature = 22
override = True

ready = trained and 15 <= temperature <= 30 or override
safe = trained and (15 <= temperature <= 30 or override)
print(ready, safe)
```

Evaluate each component and explain why the two stored results differ.

Question 2. Trace negation and a dependent readiness rule.**5 points**

```
flow_ok = True
label = "pending"
needs_review = not flow_ok or label != "approved"
ready = flow_ok and not needs_review
print(needs_review, ready)
```

Evaluate each Boolean in dependency order and give exact output.

Question 3. Program construction**10 points**

Write one complete Python program that satisfies every requirement.

- Read flow as a float, approval as text, and sensor as text.
- Print invalid if flow is negative.
- Otherwise print ready only when flow is from 18 through 22 inclusive, approval is exactly yes, and sensor is exactly ready.
- Print hold for every other nonnegative case. Print exactly one word.

Test evidence / expected result

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